

MAIE 5421 / ARIN 5201 — Assignment 1

Computer Vision (Fall 2026)

1 Simple Linear Regression

(15 Points)

Given the input variable X and the observed variable Y , derive the parameter estimates α and β for simple linear regression.

$$Y = \alpha X + \beta + \epsilon \quad \text{SSR} = \sum_{i=1}^n (y_i - (\beta + \alpha x_i))^2$$

2 Confusion Matrix

(10 Points)

Suppose you are working on a binary classification problem and obtained the following confusion matrix. Calculate the True Positive Rate (TPR), True Negative Rate (TNR), False Positive Rate (FPR), False Negative Rate (FNR), Recall, Precision, Accuracy, and F1-Score.

	Actual Positive	Actual Negative
Predicted Positive	40	5
Predicted Negative	10	45

3 K-Nearest Neighbors

(10 Points)

You are given a dataset of 10 points with three features (x, y, z) and their respective class labels A and B, shown below. Classify the point $(3.5, 4.0, 6.0)$ using K-Nearest Neighbors ($K = 3$) with two distance metrics: **L1 distance** and **cosine distance**.

Data ID	Feature x	Feature y	Feature z	Label
1	1.0	2.0	3.0	A
2	0.5	1.8	2.7	B
3	1.2	2.2	3.5	B
4	4.6	5.6	3.7	A
5	2.4	4.6	3.6	A
6	3.5	2.0	4.1	B
7	3.6	4.6	7.1	A
8	6.2	4.1	1.3	B
9	8.4	3.5	1.8	A
10	5.8	3.4	2.7	B

4 Naive Bayes Classifier

(10 Points)

Consider the observations below. A new sample is given with the attributes [Sunny, Hot, Normal, False]. Use the Naive Bayes classifier to determine the most likely decision for the given data point.

Data ID	Outlook	Temperature	Humidity	Windy	Play Golf
1	Rainy	Hot	High	True	No
2	Overcast	Hot	High	False	Yes
3	Sunny	Mild	High	False	Yes
4	Sunny	Cool	Normal	False	Yes
5	Sunny	Cool	Normal	True	No
6	Overcast	Cool	Normal	True	Yes
7	Rainy	Mild	High	False	No
8	Rainy	Cool	Normal	False	No

5 Decision Tree

(15 Points)

Given the following dataset, complete the following tasks:

- Using the ID3 algorithm, calculate the entropy of the entire dataset and the information gain for **Weather**, **Temperature**, and **Wind**.
- Using the ID3 algorithm, construct the entire decision tree. Reasoning details are required.

Data ID	Weather	Temperature	Wind	Class
1	Sunny	High	High	No
2	Overcast	High	High	Yes
3	Rain	Medium	High	Yes
4	Rain	Low	Medium	Yes
5	Overcast	Low	Medium	Yes
6	Sunny	Medium	High	No
7	Sunny	Medium	Medium	Yes
8	Rain	Medium	Medium	Yes
9	Overcast	Medium	High	Yes

6 K-Means

(10 Points)

Given the following 3-dimensional dataset containing 8 data points, use the K-Means algorithm to divide these points into 2 clusters ($k = 2$) and calculate the centroid of each cluster.

Data Point	Dim. 1 (X)	Dim. 2 (Y)	Dim. 3 (Z)
1	2.0	3.0	1.0
2	1.5	2.5	1.2
3	8.0	7.0	9.0
4	7.5	6.5	8.5
5	2.5	3.5	1.5
6	7.0	6.0	8.0
7	1.8	2.8	1.1
8	8.5	7.5	9.5

- Step 1.** Select 2 initial centroids (Data Point 1 and Data Point 3).
- Step 2.** Use Euclidean distance to calculate the distance between data points and centroids.
- Step 3.** Iteratively update the centroids for 2 rounds.

- **Step 4.** Output the centroid of each cluster and the data points it contains.

7 CNN — Parameter Counting

(15 Points)

A neural network is defined below for a classification problem. The input is an image of size $1 \times 64 \times 64$.

1. 2D convolutional layer 1: input channels 1, output channels 8, kernel 3×3 , stride 2, padding 1.
2. 2D convolutional layer 2: input channels 8, output channels 16, kernel 3×3 , stride 2, padding 1.
3. 2D convolutional layer 3: input channels 16, output channels 32, kernel 3×3 , stride 1, padding 1.
4. Max pooling layer with kernel size 2×2 .
5. Flatten layer: flatten the feature of size $32 \times 8 \times 8$ into size 2048.
6. Linear layer: input 2048 neurons, output 10 neurons.

Calculate the **total number of parameters** in the above network. Parameters include both weights and biases.

8 CNN — Forward Pass

(15 Points)

Given the following network structure:

1. **Input:** a 5×5 matrix A .
2. **Convolutional layer:** apply a 3×3 kernel K to the standardized A (no padding, stride 1, no bias).
3. **ReLU activation:** apply ReLU to the convolution results.
4. **Flatten:** flatten the result into a 1D vector.
5. **Linear layer:** pass the flattened vector through a 9×1 weight matrix W (no bias), yielding a scalar.

$$A = \begin{bmatrix} 2 & 0 & 14 & 5 & 4 \\ 11 & 12 & 16 & 4 & 8 \\ 12 & 6 & 12 & 3 & 12 \\ 10 & 10 & 2 & 20 & 11 \\ 4 & 11 & 8 & 12 & 4 \end{bmatrix} \quad K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$W = [1 \quad -2 \quad 1 \quad -1 \quad -3 \quad 0 \quad 1 \quad -1 \quad 1]^T$$

Please calculate the output.

— End of Assignment 1 —